

Liquidity Risk, Credit Risk, Market Risk and Bank Capital

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Abstract

With a sample of twelve US bond indices spanning different maturities, credit ratings and industry sectors, we investigate the impact of new bank capital regulation for trading portfolios introduced by Basel III. Specifically, we estimate the new capital requirements for (a) liquidity risk and credit risk through the so called Incremental Risk Charge, and (b) the risk of extreme market movements, which we measure with stress tests based on the 2007-2009 financial crisis. We find that capital requirements should increase substantially more than suggested by extensive impact studies conducted by the regulators with the participation of a large sample of banks. We suggest that the lower impact on capital reported by the banks may be due to the assumed risk reduction stemming from their hedging strategies. However, their effectiveness in crisis scenarios remains an open question.

Keywords: Liquidity Risk, Credit Risk, Market Risk, Financial Crisis, Basel III.

JEL Classification: G11, G21, G22, G28, G32.

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1. Introduction

In the aftermath of the 2007-2009 financial crisis bank regulators devised Basel III, a new rulebook that includes several measures to strengthen the resilience of the banking sector. A lot of effort has been put into designing new capital requirements that would provide banks with sufficient reserves to withstand future crises. As most of the losses suffered by financial institutions in the recent upheaval stemmed from their securities portfolios (Basel Committee of Banking Supervision (BCBS) 2009a), the new capital adequacy framework has put greater emphasis on the regulation of trading book risks. Specifically, the new rules focus on market risk, in normal and stressed conditions, liquidity risk and credit risk.

The literature on the effect of Basel III on bank capital is thin owing to the novelty of the regulation (preliminary versions were issued in 2009 and were only finalised in late 2010) and the fact that several of the new provisions will be phased in only in the coming years.² The Basel Committee has issued two quantitative impact studies (BCBS 2009b, 2010a) on large samples of banks across several countries. The main conclusions of the most recent and most comprehensive one are that the new requirements will cause trading book capital to increase noticeably. Specifically, the incremental risk charge (IRC) that accounts for credit risk in the trading book and the stressed VaR, that measures potential losses due to price risk in a crisis scenario, will cause the new capital requirement to go up by a median of 51.7% and 28.8% respectively.

In this study we also measure the impact of the new rules on bank capital requirements but reach very different conclusions. We find that, the size of both the IRC and stressed VaR should be much larger and may push trading book capital up to eight times higher than the pre-crisis level. As we focus on bond portfolios our results may be due to lack of diversification across different asset classes. To check this we have also considered portfolios made with stocks and bonds and found that our results hold. We conclude that diversification effects may not help to reduce the new capital requirements, especially in

² The full implementation of Basel III is not expected before the end of 2018 (BCBS 2010b, Annex 4, p. 69).

periods of market turmoil. We then argue that the lower sensitivity to the new rules reported by the banks surveyed in the Basel studies may be result of banks' recourse to hedging to reduce losses in crisis scenarios. However, given the prominent role of counterparty risk in the recent crisis and, as a result, the possibility that portfolio insurance providers may not be able to honour their commitments in a severe recession, we question whether banks' estimates may understate the potential risks.

This study contributes to the existing literature in several ways. First, we provide a simple illustration of how to account for the interaction between liquidity risk and credit risk within Basel III's incremental risk capital charge (IRC). Second, we estimate the sensitivity of credit, liquidity and market risk in trading portfolios to several risk dimensions, including credit rating, credit rating method (point-in-time or through-the-cycle), maturity and industry sector. And finally, we test the influence of portfolio diversification on the new trading book capital requirements in crisis periods.

Other studies related to Basel III have mainly focused on its macroeconomic impact. The Basel Committee (2010c) while analyzing the long term effect of the new capital rules on economic output found it to be positive. On one hand, they conclude that as higher capital requirements will make it more expensive for banks to fund their operations, the costs will be passed on to the borrowers through higher lending rates which will translate in reduced new lending activity. But, they find that the resulting downward pressure in GDP is more than compensated by the benefits of better capitalised banks which will lower the likelihood of banking crisis and of the consequent output losses. On the basis of this long term cost-benefit analysis the Committee maintain that there is room for a considerable tightening of bank capital regulation. Kashyap, Stein and Hanson (2010) support this conclusion as they find that even a 10% point increase in the capital ratio of banks will only lead to a marginal increase in lending rates of about 25 basis points. However, they argue that even small changes in the cost of borrowing in the banking sector may lead to substantial disintermediation towards the shadow banking system, which could have destabilising effects. Finally, Demirguc-Kunt, Detragiache and Merrouche (2010) measure the reaction of bank stock prices to several types of capital ratios during the financial crisis. They find evidence that an improvement in those ratios that have been

given more emphasis in Basel III (namely, leverage ratio and Tier I ratio) more positively affected stock returns.

The paper is organised as follows. In Section 2, we summarise the new regulatory setting for the trading book and describe our method to estimate the incremental risk capital charge. In Section 3, we present the data used for our analysis. Results are discussed in Section 4. Robustness tests are reported in Section 5. Section 6 concludes the paper.

2. New trading book capital requirements

2.1. Background

Banks are required to have a minimum amount of capital to be able to absorb losses and still operate as going concerns. However, during the recent crisis, the losses that banks suffered in their trading books have far exceeded minimum capital requirements (BCBS 2009c). As a result, the Basel Committee have undertaken an extensive revision of bank regulation which has resulted in several new measures (BCBS 2009c, 2010b). To increase the loss-absorbing capacity of bank capital the Basel Committee have introduced two additional capital requirements for the trading book, the “incremental risk capital” charge (IRC) and the stressed value-at-risk.

The IRC captures credit risk in (unsecuritized) trading instruments, that is the risk of losses from default and credit migration events. It was introduced to address the shortcomings of the existing value at risk framework for the trading book based on the assumption that the bank is exposed to securities price movements up to 10 trading days. The recent crisis has shown that shocks can last much longer and, due to market illiquidity, banks may be locked in their positions and unable to stop accumulating losses. The forced extension of the investment horizon leaves banks with substantial exposure to credit risk. Hence, the need for a new capital charge. The IRC is meant to measure the credit risk in a trading portfolio over a period of 1 year (capital horizon). The assumed illiquidity period (called liquidity horizon) should not be less than three months. At the end of the liquidity horizon, banks are assumed to rebalance their portfolio to reproduce

the level of risk they had at the beginning of the period.³ The logic behind this “constant level of risk” assumption is that banks, even in a crisis, will need to take on risk in order to remain profitable.⁴ Portfolio rebalancing at the end of the liquidity horizon will affect those assets that have migrated to a different credit rating. Thus, the new rules establish a relationship between migration risk and liquidity risk. This entails that close attention should be paid to the type of ratings employed to measure the probability of rating migrations. As shown by Kealhofer et al (1998) and Kealhofer (2003) there is a substantial difference in migration and default patterns between point-in-time (PIT) ratings and through the cycle (TTC) ratings. A through-the-cycle rating is typically produced by rating agencies and evaluates the performance of a company over the medium to the long-term. The objective is to arrive at a stable rating that is not affected by changes in a company’s outlook due to temporary variations in economic conditions. These types of ratings are particularly suitable for long term institutional investors. Basel regulators also favour the use of TTC ratings as they dampen the procyclicality of capital requirements.⁵ PIT ratings on the other hand, focus on the short term performance of a company. These have mostly been used by banks as they are interested in the ability of a firm to repay its loans, which are typically short term. Interestingly, in order to provide ratings that are more in line with current conditions - also following criticism for TTC ratings’ inaccuracy during the South-East Asian crisis and more recently the Enron and Worldcom debacles and the subprime crisis - rating agencies have now started to provide PIT ratings alongside TTC ones.⁶ For an analysis of the properties of both types of ratings see, for example, Kealhofer et al (1998), Carey and Treacy (2000), Carey and Hrycay (2001), Kealhofer (2003) and Kou and Varotto (2008). In this work, we shall explore the different impact of PIT and TTC ratings on the incremental risk charge.

The IRC should be estimated with a VaR model with a 99.9% confidence level which is in line with the VaR parameterization employed in the banking book (i.e. the loan book) under the internal rating based approach (IRBA). This way the IRC should achieve the

³ BCBS (2009a), p. 3.

⁴ BCBS (2009a), footnote 3, p. 3.

⁵ See, for example, Resti and Sironi (2007), p. 370-372.

⁶ For instance, Moody’s recently purchased KMV corporation which specialised on PIT ratings. Moody’s now provides Moody’s-KMV ratings together with their traditional TTC ratings.

objective of harmonising the treatment of credit risk in both the trading book and banking book, and thus reduce the scope of banks choosing strategically to allocate assets to one book or the other to lower their regulatory capital.

During the recent crisis, securities prices have changed considerably across markets and types of financial instrument. Although credit risk (in the form of default and migration risk) has undoubtedly played a role,⁷ a lot of the price variation was likely related to market risk factors, such as changes in risk premia (Berg 2010). This conclusion is also consistent with the finding in Elton et al (2001) and Gieseke et al (2009) who show how risk premia may have a larger effect on bond returns than default risk factors. To address this point, the Basel Committee has introduced, on top of the IRC, another capital add-on designed to make bank capital able to absorb sharp negative price changes in a crisis. Price risk is measured with a value-at-risk model estimated under stressed market conditions.

To arrive at the total capital for the trading book, banks will need to add the IRC and stressed VaR to the current VaR of their trading portfolios. If the internal VaR model used by the bank does not capture firm specific risks, then those will also need to be added separately (BCBS 2009c, p. 18, paragraph 718 LXXXVII-1-). In summary, the old and new capital requirements will be given by,

Old capital requirement = Current VaR + Specific risk charge

New capital requirement = Current VaR + Specific risk charge + IRC + Stressed VaR

A separate add-on for specific risk will be necessary when banks compute the VaR with factor models that only capture the systematic risk component of returns. The specific risk component may not be modelled on the assumption that it is diversified away. In our analysis we shall not employ factor models but compute VaR directly on the empirical

⁷ When looking at the universe of Moody's rated issuers, corporate default rates, across all ratings, have climbed from 0.37% in 2007 to 2.02% in 2008 to a 76 year high of 5.36% in 2009 which is the third largest default rate ever recorded since 1920 and was exceeded only during the Great Depression in 1933 (8.42%) and 1932 (5.43%). For details, see Moody's (2010), Exhibit 31, pp. 42-44.

return distribution of the portfolios we consider. As a result, any residual specific risk will be already accounted for in the “Current VaR”.

2.2. The IRC

We estimate the IRC with the IRBA model in Basel II. This is consistent with the Basel Committee requirement that “[t]he bank must demonstrate that the approach used to capture incremental risks [i.e. the IRC] meets a soundness standard comparable to that of the IRBA approach for credit risk ...”.⁸ The IRBA capital requirement $K_{z,g}$ for a corporate exposure z with rating g is,

$$K_{z,g} = CF \cdot MA_{z,g} \cdot \left[(1 - a_z) P_{D,1,g} - (1 - a_z) P_{A,1,g} \right] \cdot EAD_z \quad (1)$$

where CF is a calibration factor, currently equal to 1.06, which was introduced by the Basel Committee to make the IRBA deliver, on average across the banking sector, a similar level of capital to that resulting under the previous regime, i.e. Basel I;⁹ $MA_{i,g}$ is a “maturity adjustment” that increases with the exposure’s “effective” maturity¹⁰ to reflect the higher potential for downgrade risk, and ultimately default risk, of longer maturity assets. $P_{D,1,g}$ is the 1-year probability of default under a downturn scenario, whereas $P_{A,1,g}$ is an average, or through-the-cycle, 1-year default probability. After estimating $P_{A,1,g}$ with historical data, $P_{D,1,g}$ is obtained by rescaling $P_{A,1,g}$ upward with a deterministic expression.¹¹ EAD_z is the exposure at default for asset z ; and a_z is the recovery rate. For more details about (1) see, for example, Resti and Sironi (2007)¹² and BCBS (2005). Note that the term $(1 - a_z) P_{D,1,g}$ represents the expected loss in a

⁸ BCBS (2009c), statement 718(XCIII), p. 20

⁹ BCBS (2006), paragraph 14, page 4.

¹⁰ Effective maturity in Basel II, is measured as a Macaulay duration under the assumption that interest rates are zero.

¹¹ BCBS (2005).

¹² See Section 20.4, pp. 597-612.

downturn for exposure z and $(1 - a_z)P_{A,1,g}$, the exposure's average expected loss. The difference between these two terms is called unexpected loss, and constitutes the basis for the calculation of credit risk capital of most credit risk models employed in the industry. The idea is that capital should only cover for unexpected events in that banks already set aside loan provisions against expected losses.

The IRC applies to those exposures that are “subject to a capital charge for specific interest rate risk, with the exception of securitization exposures and n-th-to-default credit derivatives”.¹³ The IRC requires “banks that model specific risk to measure and hold capital against default risk that is *incremental* to any default risk captured in the bank's value-at-risk model.”¹⁴ The IRC “should be based on the assumption of a constant level of risk under the one-year capital horizon”.¹⁵ As mentioned before, this means that the portfolio is rebalanced to preserve its initial credit rating profile and concentration level. The rebalancing period can be exposure specific. The higher the concentration of the portfolio on one exposure the longer should be its liquidity horizon. Therefore, the liquidity horizon can be used to account for both liquidity risk as well as concentration risk.

The constant risk rule implies that migration risk (i.e. upgrades and downgrades to non-default states) takes place only within the liquidity horizon. Default risk, on the other hand, will always be estimated over the capital horizon of one year. Then rebalancing at the end of the liquidity horizon will cause the risk profile of the portfolio to be realigned several times every year. As a result, annual default rates will be affected. We use the following procedure to measure the impact of the liquidity horizon on one-year default probabilities:

1. We estimate the “generator matrix” Q from a one-year transition matrix $M_{0,1}$ where the first subscript indicates the beginning of the transition period and the second one the end, when time is measured in years. The generator is a useful

¹³ BCBS (2009c), statement 718(XCII), p. 20.

¹⁴ BCBS (2009a), p.1.

¹⁵ BCBS (2009a), p.3.

tool as it allows one to obtain transition matrices over any desired time period (for details see, Jarrow et al, 1997). To estimate the generator we follow the procedure illustrated in Israel, Rosenthal and Wei (2001),

$$Q = \sum_{h=1}^{\infty} \frac{(M_{0,1} - I)^h}{h} (-1)^{h+1} \quad (2)$$

where I is an identity matrix. As h increases, convergence is achieved quickly, so the derivation of the generator does not normally pose computational problems.

2. From the generator, we produce a transition matrix over the desired liquidity horizon, say, three months. Let us denote this matrix as $M_{0,0.25}$. Then,

$$M_{0,0.25} = I + \sum_{h=1}^{\infty} \frac{Q^h 0.25^h}{h!} \quad (3)$$

As h increases the above summation converges quickly, which implies that the derivation of $M_{0,0.25}$ is straightforward. The default probabilities $P_{A,0.25,g}$ in the default column of the resulting matrix will then incorporate the effect of migration risk over the liquidity horizon.

3. We then derive one-year default probabilities $P_{A,1,g}^*$ from the default probabilities obtained in the previous Step under the assumption that the portfolio is rebalanced at the end of each liquidity horizon (i.e. every three months). This can be done by cumulating over one year the default probabilities in Step 2 without allowing for migration risk from one quarter to the next,

$$P_{A,1,g}^* = \sum_{v=0}^3 (1 - P_{A,0.25,g})^v P_{A,0.25,g} \quad (4)$$

The above equation states that the cumulative one-year default probability is

given by the probability of default in the first three-month period $P_{A,0.25,g}$ plus the probability of default in each of the other quarters when no default has occurred in any of the previous quarters, $(1 - P_{A,0.25,g})^v P_{A,0.25,g}$.¹⁶

4. In the last step, we derive the IRC capital add-on with (1). Now, however, historical one-year default probabilities $P_{A,1,g}$ are replaced with $P_{A,1,g}^*$. This approach is consistent with the method suggested by the Committee of European Banking Supervisors (2009). We shall use two reference transition matrices, a through-the-cycle (TTC) matrix and a point-in-time (PIT) matrix, which are characterised by different levels of migration risk. These are reported in Table 1. $P_{A,1,g}^* / P_{A,1,g}$ ratios for different liquidity horizon and transition matrix assumptions are shown in Table 2.

TTC transition matrices are based on TTC ratings that only migrate when a change in the underlying credit quality of the rated entity is considered to be permanent. An example of TTC matrix is the Moody's 1920-2008 average transition matrix in Table 1 Panel A from Moody's (2009). PIT transitions, on the other hand, are based on PIT ratings. PIT ratings are more volatile, i.e. tend to migrate more, because they quickly adjust to reflect current changes in credit quality, whether of permanent or temporary nature. In Table 1 Panel 2 we report a sample PIT transition matrix estimated by KMV with company data from 1990 to 1995.¹⁷ By comparing the two matrices in the Table it is immediately clear that PIT transitions to non-default states are far larger, and the probabilities of no-migration far smaller, than in the TTC case. In Table 2 we quantify the influence of liquidity horizons of 1, 3, 6 and 12 months and different transition matrices on one-year default probabilities. The one-year default probabilities typically fall as the liquidity horizon gets shorter due to the declining impact of downgrade risk. This holds true for all investment grade ratings. On the other hand, for the worst rating category (CCC) in which, if default

¹⁶ Equation (4) can be obtained by multiplying four times by itself the three month transition matrix $M_{0,0.25}$ computed in Step 2, after setting its transition probabilities to non-default states to zero and the probabilities in the main diagonal to $(1 - P_{A,0.25,g})$. $P_{A,1,g}^*$ will then be the cumulative default probability of rating g found in the default column of the resulting matrix.

¹⁷ See Gupton et al (1997), p. 70.

does not occur, credit migration can bring a substantial improvement in credit quality, shorter default horizons will increasingly eliminate this substantial upside and, as a result, bring steadily rising one-year default rates. A similar pattern was observed by Dunn et al (2006). Results for intermediate categories are more ambiguous (see Panel B).

Substantial differences in the sensitivity to the liquidity horizon are found for PIT and TTC ratings. A comparison between the two Panels in Table 2 reveals that, for investment grade assets, shorter liquidity horizons bring about a greater reduction in one-year default rates with PIT ratings. This is because a greater proportion of default events are preceded by downgrades when PIT ratings are used as opposed to TTC ratings. Then, through portfolio rebalancing, downgraded assets are replaced with higher quality ones at a faster rate with PIT ratings as the liquidity horizon shortens. On the contrary, for the lowest rating category, decreasing migration risk via portfolio rebalancing reduces the potential for upgrades to a greater extent with PIT ratings than with TTC ratings. This results in default rates being higher at shorter liquidity horizons. Again, the change is more pronounced for PIT ratings. The greatest reduction in one-year default rates are found for AAA assets over a 1 month liquidity horizon, which fall by 69.3% when compared with a 12 month liquidity horizon. The largest increase is by 10.8% for CCC assets over a 1-month liquidity horizon. The largest fall (increase) with TTC ratings is for BBB (CCC) by 17.6% (3.2%) over a 1 month liquidity horizon.¹⁸

2.3. The Stressed VaR

The newly proposed stressed value-at-risk¹⁹ (VaR^{stress}) should be based on a 10 day VaR with a 99% confidence interval and estimated with one year of historical data from a

¹⁸ It may be argued that the difference between Panel A and B in Table 2 may be due to the fact that PIT and TTC transition matrices differ both in terms of migration probabilities *as well as* default probabilities. We have therefore re-computed Panel B results (not reported) by replacing the default probabilities of the PIT matrix with those of the TTC matrix. Diagonal elements of the PIT matrix have been adjusted to ensure that the sum of the probabilities in each row of the matrix is equal to 1. The new matrix, which differs from that in Panel A only for higher migration risk, but has identical default risk, produces the same patterns observed when comparing Panel A and B.

¹⁹ BCBS 2009c, par. 718(lxxvi), p. 13-15.

continuous period of significant stress. The old capital requirement, on the other hand, is based on a 10 day 99% VaR calculated with the most recent data. For both VaRs the capital requirement on a particular day t will be the highest of the previous day's VaR and the average VaR over the previous 60 days multiplied by a scaling factor. The scaling factors take values between 3 and 4 depending on the historical performance, commonly called “backtesting”, of the two VaR models²⁰.

Previous literature has looked at the accuracy of VaR and whether it should be used for regulatory capital purposes. Recently, Perignon et al (2010a, b) conclude that VaR is often inaccurate and has little ability to predict future portfolio outcomes. With pre-crisis data they find that banks systematically overestimate their trading book capital needs. However, the current crisis has painfully highlighted that banks were heavily undercapitalised and that their trading book VaRs were grossly understating risk. This was likely the case because the pre-crisis period, on which VaR models were estimated, was characterised by unusually low volatility (Acharya and Schnabl, 2009). In addition, Jimenez-Martin et al (2009) conclude that “[t]he concept of VaR is intended to capture possibly bad outcomes on a typical day, and not when market panic sets in.” Based on these arguments, a stressed VaR, estimated over a past period of severe market distress, appears to be justified as an additional measure to correct for the potential capital shortfalls that can result when employing unstressed VaRs.

3. The data

To compare the size of new and old capital requirements in the trading book we estimated the IRC, the pre-crisis VaR and the stressed VaR for twelve bond portfolios with different credit rating and maturity characteristics. The portfolios are represented by US corporate bond indices compiled by Bank of America-Merrill Lynch and sourced from Datastream. The indices span two industry sectors, industrial and financial, two rating groups, AAA-AA and A-BBB, and three maturity bands, 5 to 10 years, 10 to 15 years and 15+ years. The sample consists of daily returns over the period May 2004 – August 2009. The

²⁰ For more information on backtesting see Basel (1996b). For a criticism of the backtesting procedures implemented under Basel 2 see Jimenez-Martin, McAleer and Perez-Amaral (2009) and McAleer (2009).

period was chosen to include the recent crisis and to allow for enough observations to determine the pre-crisis VaR. Summary statistics are reported in Table 3. All indices exhibit negative skewness (except for the AAA-AA 10-15 year maturity index) and substantial excess kurtosis. Financial indices are almost invariably more negatively skewed and have always fatter tails than industrial indices.

4. Results

To implement the model and calculate the IRC for our sample of corporate bond indices we need to estimate two parameters, the recovery rate a_z and the average 1-year default probability adjusted to account for a 3-month liquidity horizon, $P_{A,1,g}^*$. All the other parameters in the model are deterministically related to the default probability and bond maturity and hence will be automatically determined once the latter are known. In Section 2.1. we have described how $P_{A,1,g}^*$ can be estimated starting from a 1 year transition matrix. We employ the TTC and PIT transition matrices reported in Table 1 to obtain two estimates of $P_{A,1,g}^*$ for each rating grade g (and given maturities). Lacking information on the average coupon of the bond indices in our sample we shall assume that the indices are made of par coupon bonds.²¹ Robustness tests conducted with alternative coupon assumptions indicate that results would vary only marginally (see Section 5). It should be noted that the IRC estimates reported in Table 4 are almost unchanged across the three maturity bands considered: 5-10 years, 10-15 years and 15+ years. This is because the IRBA model in (1) was designed by the Basel Committee to be sensitive to changes in (effective) maturity from 1 to 5 years but not beyond 5 years, and we only consider bond indices with maturities of 5 years and above. Finally, according to Basel regulation a_z should be a “downturn” recovery rate and should be employed to compute the downturn expected loss as well as the average expected loss in (1).²² It may be argued that adopting the same downturn recovery to estimate both losses may not be

²¹ To derive the par coupons we first determine the bonds’ theoretical prices based on their credit rating and maturity. We employ the risk neutral pricing suggested in Elton et al (2001).

²² BCBS (2005).

appropriate. Therefore, we have also computed the IRC when average losses are estimated with an average recovery across the business cycle but found only minor changes in our results. These are discussed in Section 5.

In Table 4 we compare the IRC with (a) the value-at-risk based trading book capital requirements estimated before the 2007-2009 crisis²³, and (b) with the newly introduced stressed value-at-risk, which is the highest VaR over the crisis period.²⁴ We do so in order to evaluate the impact of the current proposals on the total capital requirement for the trading book. We estimate pre-crisis VaR and stressed VaR with the set of bond indices described in Section 3. Then, the two definitions of capital we will compare are,

Old capital requirement = Pre-crisis VaR

New capital requirement = Pre-crisis VaR + IRC + Stressed VaR

We find that,

- i. As expected, the size of the IRC capital depends on the rating method employed in the analysis. With through-the-cycle ratings the IRC ranges from 4.91% of the total exposure for 5-10 year AAA-AA indices and 8.71% for 15+ year A-BBB indices. With point-in-time ratings, the percentages are lower at 3.85% and 7.39% respectively, due to the lower default rate of such ratings. IRC values across the financial and industrial sectors are identical because we could not source sector specific transition matrices (and default rates) for PIT ratings. So, for consistency, we adopted a unique transition matrix (and a unique set of default rates) across all sectors for both types of rating. However, we capture industry effects in the “stressed” value-at-risk which is discussed below.

²³ One of the earliest signs of the crisis was the failure of Bear Sterns to provide financial support to one of its hedge funds on 22nd June 2007. We estimate pre-crisis VaR with data up to 20th June 2007.

²⁴ We compute the value at risk by taking the difference between the mean and the 1% quantile of the empirical distribution of daily portfolio returns. Each day, the distribution is obtained from the past 250 observations. To derive the 10-day VaR we multiply the daily VaR by the square root of 10, a common procedure that complies with the regulatory framework for market risks set out in the 1996 Amendment (BCBS 1996a, p. 44).

- ii. The size of the IRC is substantial relative to the old trading book capital requirement but it varies markedly across ratings and maturities. This is primarily because old capital is very sensitive to these risk dimensions. The ratios of IRC to old capital range from a minimum of 39% (AAA-AA 15+ year industrial index with PIT ratings) to 173% (A-BBB 5-10 year industrial index with TTC ratings).
- iii. The stressed VaR is always significantly larger than the IRC. For industrial bonds, stressed VaR ranges from a minimum of 1.98 times the level of the IRC (for A-BBB bonds with 5-10 year maturity with TTC ratings) and a maximum of 6.82 times (for 10+ year AAA-AA bonds with PIT ratings). The ratios between the two risk measures are even larger for financial bonds where they range from 3.12 to 10.11.
- iv. As a result of the previous observations, the newly proposed capital requirements for the trading book are much bigger than the old capital requirements. The reported New/Old capital ratios allow us to illustrate this point and show that new capital ranges, for industrial bonds, from a minimum of 4.07 times old capital (AAA-AA 15+ maturity bonds with PIT ratings) – which corresponds to an increase of 307% - to a maximum of 6.17 times (A-BBB 5-10 year maturity bonds with TTC ratings) – an increase of 517%. The multiples are bigger for financial bonds and are 5.48 (448%) and 9.07 (807%) respectively. In other words, following the current crisis, banks may be required to increase their trading book capital by more than 8 times relative pre-crisis levels, on the basis of the new rules.

The Basel Committee has recently completed two quantitative impact studies (QIS2009 and QIS2010) on the new proposals for computing trading book capital requirements (BCBS 2009b, BCBS 2010a). The findings reported by 43 participating banks across 10 countries in QIS2009 show that the median increase in trading book capital would be 60.4% due to the IRC (for a 3 month liquidity horizon) and 63.2% as a result of the new stressed VaR requirement. Standard deviations around the average are substantial (125.1% and 130.8% respectively). In the QIS2010 which was extended to 94 large, well

diversified and internationally active banks from 23 countries, the median increase in trading book capital due to the IRC and the stressed VaR is lower at 28.8% and 51.7% respectively with lower standard deviations (49.1% and 43.8%). Our results differ markedly from those reported by the Committee. While we find that both the IRC and stressed VaR are higher than in the Basel studies, the difference is particularly pronounced for the stressed VaR. As a result, our total capital increase following the introduction of the new rules is also much larger. A possible explanation may be that in this study we focus on interest rate risk instruments while a typical bank's trading book may include exposures to other types of risk, that is equity, commodity and foreign exchange risk. The resulting diversification effects may decrease the stressed VaR and hence the total capital of the bank under the new rules. However, in a severe crisis like the one began in 2007, which affected virtually all asset classes at the same time, diversification may not offer real benefits. We have tested this idea by constructing mixed portfolios of stocks and bonds. To do so, we have used the 12 bond indices employed in the previous analysis and combined them with the S&P500 stock index. We have then computed the old trading book capital of the diversified portfolios (that is their pre-crisis VaR) and their stressed VaR as before. Results are reported on Table 5. The capital increase that can be attributed to the stressed VaR varies in relation to the composition of the mixed portfolios but, by and large, its magnitude is comparable to an undiversified bond portfolio and far greater than shown in the quantitative impact studies.²⁵ In conclusion, our analysis begs the question of how banks can achieve such substantially lower stressed VaR under the new rules. An explanation could be that banks' portfolios are now heavily hedged against crisis events. Indeed, following the recent market dislocation, and the large losses suffered by many banks as a consequence, hedging is certainly a rational response. However, one may wonder whether such hedges would hold in a severe crisis and to what extent counterparty risk would reduce their effectiveness if a crisis were to materialise. If banks are overconfident about their risk management skills and their ability to cope with future shocks then, clearly, the introduction of the new rules would not be as effective as intended.

²⁵ In the Basel studies the pre-crisis VaR was estimated as of 31st December 2006 rather than 20th June 2007 as in this work. We have re-done our analysis with the Basel definition of pre-crisis period but found no meaningful change in our results.

5. Robustness tests

To derive the IRC we have made the following assumptions which represent our “benchmark” case: (a) the recovery rate does not change with the business cycle, (b) the risk free term structure is flat at 0% and (c) the indices under study are made of par coupon bonds. In this Section we present robustness tests to check the sensitivity of our results to these assumptions. Although assumptions (a) and (b) are consistent with the internal rating based approach to computing credit risk capital, they appear to be too restrictive. After all, interest rates may significantly depart from zero and recoveries are known to be positively related with the business cycle. The results of the tests are reported in Table 6. The table shows how the IRC based on through-the-cycle and point-in-time ratings vary when interest rates increase from 0% (Panel A) to 3% (Panel B) and 6% (Panel C). Within each panel we then show how the IRC changes when from a common downturn recovery of 21% used to derive the unconditional expected credit loss and the downturn expected credit loss in equation (1), we consider an average recovery of 45% for the average loss. The average and downturn recoveries have been estimated by taking, respectively, the mean and the minimum historical recoveries in the 1982-2008 sample for senior unsecured bonds provided by Moody’s Investors Service (Moody’s 2009). Within each panel we also consider zero coupon bonds in alternative to par coupon bonds. In Panel A we can see that when we depart from the benchmark case by switching to zero coupon bonds the maximum IRC change is for A-BBB 15+ year maturity bond indices with TTC ratings and amount to only 4 basis points from 8.71% to 8.67%. If we keep par coupon bonds but differentiate between downturn and average recoveries the largest IRC move would be for A-BBB bonds (across all maturities) and equal 10 basis points. This is because the IRC is mainly driven by the downturn default probability in the downturn credit loss. So the recovery employed to derive the average credit loss is almost inconsequential. Finally, the largest change in IRC occurs when we increase the interest rates. When we rise them from 0% to 6% for instance, all else equal (i.e. while we retain a par coupon and constant recovery assumptions), the maximum IRC increase will be of 52 basis points from 8.70% to 9.22% for A-BBB bonds with 10-15 year maturity and with TTC ratings. Overall, our tests indicate that our results and conclusions are robust to changes in our main assumptions.

6. Conclusion

In this paper, we explore the impact of new rules that will require banks to hold capital reserves against market, credit and liquidity risk in their trading books. We find that the new requirement for incremental credit risks, the IRC, may be substantial when compared with old regulatory capital levels. However, stressed VaRs estimated on corporate bond portfolios during the current crisis show that market risk related losses may be far greater and, depending on the characteristics of the portfolio, be more than ten times as large as the IRC. Our findings are at odds with evidence presented by the Basel Committee. Our results show a much greater increase in capital following the introducing of the new rules. We conjecture that banks may be able report lower sensitivity to the new regulation because of the risk reductions they may achieve through hedging. However, if this was the case, one may question the extent to which hedging counterparts may be able to fulfil their contractual obligations in the case of a shock such as the one observed in the recent crisis. If counterparty risk was substantial, then the efficacy of the new bank capital adequacy rules may be greatly impaired by the assumptions that banks make about their ability to manage risk in extreme scenarios. Future research should aim to shed more light on this important issue. Further investigation should also be directed to the influence that the stressed VaR requirement will have on the pricing of tradeable assets as well as its implications for banks' asset allocation strategies. For instance, if following the introduction of the new rules, trading book instruments like stocks and bonds become too capital intensive relative to bank loans, then banks may prefer to redirect their investments to the latter type of assets. The result could be a move towards more traditional banking which would counter the trend observed in the last decades and culminated with the repeal of the Glass and Steagal act in 1999. The implications for banks' profitability, availability of credit, financials stability and economic growth may be substantial and deserve further research.

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Table 1: TTC and PIT Transition Matrices

In this Table we report an annual through-the-cycle (TTC) transition matrix estimated by Moody's over the sample period from 1920 to 2008 and an annual point-in-time (PIT) transition matrix estimated by KMV with data from 1990 to 1995. All figures are in percent.

Panel A: Moody's through-the-cycle transition matrix, 1920-2008, %

	Aaa	Aa	A	Baa	Ba	B	Caa-C	D
Aaa	91.1	7.8	0.9	0.2	0.0	0.0	0.0	0.0
Aa	1.3	90.6	7.1	0.7	0.2	0.0	0.0	0.1
A	0.1	3.2	90.2	5.6	0.7	0.1	0.0	0.1
Baa	0.0	0.3	5.0	87.9	5.4	0.8	0.2	0.3
Ba	0.0	0.1	0.5	6.6	83.0	7.5	0.7	1.5
B	0.0	0.1	0.2	0.7	6.8	81.6	6.4	4.3
Caa-C	0.0	0.0	0.1	0.1	0.7	6.5	74.4	18.2

Panel B: KMV's point-in-time transition matrix, 1990-1995, %

	AAA	AA	A	BBB	BB	B	CCC	D
AAA	66.3	22.2	7.4	2.5	0.9	0.7	0.1	0.0
AA	21.7	43.0	25.8	6.6	2.0	0.7	0.2	0.0
A	2.8	20.3	44.2	22.9	7.4	2.0	0.3	0.1
BBB	0.3	2.8	22.6	42.5	23.5	7.0	1.0	0.3
BB	0.1	0.2	3.7	22.9	44.4	24.5	3.4	0.7
B	0.0	0.1	0.4	3.5	20.5	53.0	20.6	2.0
CCC	0.0	0.0	0.1	0.3	1.8	17.8	69.9	10.1

Table 2: One-Year Default Probabilities and Liquidity Horizons

This table shows the influence of the portfolio rebalancing period, called liquidity horizon, on annual default probabilities. Default probabilities obtained from different rating models, through-the-cycle and point-in-time, are reported in Panel A and B respectively. Rebalancing is conducted at the end of the liquidity horizon to maintain the risk profile of the portfolio at the beginning of the period. The capital horizon is the period over which regulatory capital is computed under the incremental risk capital charge (IRC). All the default probabilities in the Table are defined over 1 year to conform with the capital horizon in the IRC.

	Months			
Capital horizon	12	12	12	12
Liquidity horizon	1	3	6	12

**Panel A: Through-the-Cycle transition matrix:
Moody's average 1920-2008**

Credit Rating	Default Probabilities, %			
AAA	0.00	0.00	0.00	0.00
AA	0.07	0.07	0.07	0.07
A	0.07	0.07	0.07	0.08
BBB	0.25	0.26	0.28	0.31
BB	1.37	1.39	1.42	1.48
B	3.86	3.94	4.07	4.29
CCC	18.79	18.68	18.53	18.21
Ratios relative to 12 month capital horizon and 12 month liquidity horizon, in percent				
AAA	na	na	na	na
AA	96.17	96.80	97.80	100
A	83.15	86.09	90.61	100
BBB	82.43	85.63	90.44	100
BB	92.54	93.86	95.89	100
B	89.96	92.01	94.89	100
CCC	103.21	102.63	101.76	100

**Panel B: Point-in-Time transition matrix:
KMV 1990-1995**

	Default Probabilities, %			
AAA	0.01	0.01	0.01	0.02
AA	0.01	0.02	0.03	0.04
A	0.06	0.07	0.08	0.10
BBB	0.12	0.16	0.20	0.26
BB	0.72	0.70	0.71	0.71
B	0.74	1.04	1.42	2.01
CCC	11.23	10.96	10.60	10.13
Ratios relative to 12 month capital horizon and 12 month liquidity horizon, in percent				
AAA	30.66	38.92	55.77	100
AA	33.96	47.51	67.00	100
A	63.88	67.25	75.95	100
BBB	47.07	59.94	76.11	100
BB	100.81	98.38	99.75	100
B	36.85	51.83	70.51	100
CCC	110.82	108.23	104.61	100

Table 3: Bond Index Summary Statistics

The Table shows summary statistics for the daily percentage returns of 12 bond indices of different credit quality, maturity and industry sector. The sample period is 7/5/2004-21/08/2009.

	Bond Indices					
	AAA-AA			A-BBB		
	5-10y	10-15y	15+y	5-10y	10-15y	15+y
	Industrials					
Mean	0.03	0.03	0.03	0.02	0.02	0.02
Volatility	0.39	0.49	0.70	0.37	0.43	0.64
Skewness	-0.05	-0.18	-0.04	-0.34	-0.55	-0.19
Excess Kurtosis	4.74	6.24	2.69	5.47	4.92	2.61
Minimum	-2.34	-3.73	-3.39	-2.26	-3.24	-4.13
Maximum	2.82	3.32	3.67	2.65	2.28	2.90
	Financials					
Mean	0.02	0.02	0.02	0.01	0.01	0.01
Volatility	0.50	0.71	0.80	0.54	0.54	0.71
Skewness	-1.94	0.58	-1.02	-2.00	-0.61	-1.42
Excess Kurtosis	40.34	31.44	12.61	29.39	8.72	13.93
Minimum	-6.56	-7.06	-8.31	-6.44	-4.33	-7.94
Maximum	4.49	8.42	3.87	3.94	3.59	3.10

Table 4: New and Old Trading Book Capital Requirements

The table shows capital charges as percentages of exposure value. Sample period: 7/5/2004-21/08/2009. PIT and TTC stand for "point-in-time" and "through-the-cycle" respectively. Old capital is given by the pre-crisis VaR estimated with data before 20/6/2007. Stressed VaR is the maximum VaR over the sample period. The new capital is given by pre-crisis VaR plus stressed VaR plus IRC.

	Bond Indices					
	AAA-AA			A-BBB		
	5-10y	10-15y	15+y	5-10y	10-15y	15+y
	Industrials, %					
Old Capital	5.48	6.31	9.82	5.01	6.44	9.28
Stressed VaR (SVaR)	17.25	22.13	26.30	17.20	18.87	22.23
SVaR/Old Capital	3.15	3.51	2.68	3.43	2.93	2.40
	3 Month Liquidity Horizon and TTC Ratings					
IRC	4.91	4.92	4.92	8.69	8.70	8.71
IRC/Old Capital	0.90	0.78	0.50	1.73	1.35	0.94
SVaR/IRC	3.51	4.50	5.35	1.98	2.17	2.55
New Capital	27.64	33.36	41.03	30.89	34.00	40.21
New/Old Capital	5.05	5.28	4.18	6.17	5.28	4.33
	3 Month Liquidity Horizon and PIT Ratings					
IRC	3.85	3.86	3.86	7.37	7.39	7.39
IRC/Old Capital	0.70	0.61	0.39	1.47	1.15	0.80
SVaR/IRC	4.47	5.74	6.82	2.33	2.55	3.01
New Capital	26.58	32.30	39.97	29.58	32.69	38.90
New/Old Capital	4.85	5.12	4.07	5.90	5.08	4.19
	Financials, %					
Old Capital	5.43	7.76	9.57	5.39	6.50	8.61
Stressed VaR (SVaR)	34.18	35.12	39.00	34.80	27.10	31.61
SVaR/Old Capital	6.29	4.52	4.07	6.46	4.17	3.67
	3 Month Liquidity Horizon and TTC Ratings					
IRC	4.91	4.92	4.92	8.69	8.70	8.71
IRC/Old Capital	0.90	0.63	0.51	1.61	1.34	1.01
SVaR/IRC	6.96	7.14	7.93	4.01	3.12	3.63
Total New Capital	44.53	47.80	53.49	48.88	42.31	48.93
New/Old Capital	8.20	6.16	5.59	9.07	6.50	5.68
	3 Month Liquidity Horizon and PIT Ratings					
IRC	3.85	3.86	3.86	7.37	7.39	7.39
IRC/Old Capital	0.71	0.50	0.40	1.37	1.14	0.86
SVaR/IRC	8.87	9.11	10.11	4.72	3.67	4.28
Total New Capital	43.47	46.74	52.43	47.57	40.99	47.61
New/Old Capital	8.00	6.02	5.48	8.83	6.30	5.53

Table 5: Stressed VaR in Stock-and-Bond Portfolios

The table shows capital charges as percentages of exposure value for various portfolios of stocks and bonds. Sample period: 7/5/2004-21/08/2009. Old capital is given by the pre-crisis VaR estimated with data before 20/6/2007. Stressed VaR is the maximum VaR over the sample period.

Portfolio Composition: 75% in the S&P500 and 25% in each of the following bond indices						
	AAA-AA			A-BBB		
	5-10y	10-15y	15+y	5-10y	10-15y	15+y
Industrials, %						
Old Capital	12.92	12.97	13.33	12.95	13.05	13.44
Stressed VaR	60.05	58.63	58.90	60.75	60.69	60.14
Stressed VaR/Old Capital	4.65	4.52	4.42	4.69	4.65	4.47
Financials, %						
Old Capital	12.97	12.64	13.42	12.98	13.02	13.44
Stressed VaR	59.91	59.87	59.24	61.02	59.62	58.82
Stressed VaR/Old Capital	4.62	4.74	4.41	4.70	4.58	4.38
Portfolio Composition: 50% in the S&P500 and 50% in each of the following bond indices						
Industrials, %						
Old Capital	8.78	8.79	9.56	8.91	8.88	9.70
Stressed VaR	37.82	37.03	34.21	38.88	38.80	35.92
Stressed VaR/Old Capital	4.31	4.22	3.58	4.37	4.37	3.70
Financials, %						
Old Capital	8.80	9.06	9.57	8.84	9.12	9.52
Stressed VaR	40.22	37.32	36.37	41.54	37.21	38.84
Stressed VaR/Old Capital	4.57	4.12	3.80	4.70	4.08	4.08
Portfolio Composition: 25% in the S&P500 and 75% in each of the following bond indices						
Industrials, %						
Old Capital	5.48	6.11	8.52	5.40	5.77	7.75
Stressed VaR	17.24	20.76	22.04	20.19	24.14	25.67
Stressed VaR/Old Capital	3.15	3.40	2.59	3.74	4.18	3.31
Financials, %						
Old Capital	5.51	6.97	8.26	5.53	5.93	7.40
Stressed VaR	27.69	27.08	32.53	32.40	26.86	27.42
Stressed VaR/Old Capital	5.03	3.88	3.94	5.86	4.53	3.71

Table 6: IRC Robustness Tests

The table reports the IRC computed under alternative assumptions. The benchmark case employed to derive the results in Table 4 is highlighted in yellow. EL denotes unconditional expected loss, ELD expected loss in a downturn, TTC through-the-cycle ratings and PIT point-in-time ratings. The downturn recovery is the lowest recovery for senior unsecured bonds in the period 1982 to 2008 and equals 21%. The average recovery is the average over the same period and equals 45%.

Bond Indices						
	AAA-AA			A-BBB		
	5-10y	10-15y	15+y	5-10y	10-15y	15+y
Panel A: 0% risk free rates						
	Par coupon - downturn recovery for both EL and ELD					
TTC	4.91	4.92	4.92	8.69	8.70	8.71
PIT	3.85	3.86	3.86	7.37	7.39	7.39
	Zero coupon - downturn recovery for both EL and ELD					
TTC	4.91	4.91	4.91	8.67	8.67	8.67
PIT	3.85	3.85	3.85	7.36	7.36	7.36
	Par coupon - downturn recovery for ELD and average recovery for EL					
TTC	4.95	4.95	4.96	8.78	8.80	8.80
PIT	3.88	3.88	3.88	7.45	7.46	7.46
	Zero coupon - downturn recovery for ELD and average recovery for EL					
TTC	4.95	4.95	4.95	8.76	8.76	8.76
PIT	3.88	3.88	3.88	7.43	7.43	7.43
Panel B: 3% risk free rates						
	Par coupon - downturn recovery for both EL and ELD					
TTC	5.03	5.06	5.06	8.89	8.96	8.96
PIT	3.94	3.97	3.97	7.55	7.61	7.61
	Zero coupon - downturn recovery for both EL and ELD					
TTC	4.91	4.91	4.91	8.67	8.67	8.67
PIT	3.85	3.85	3.85	7.36	7.36	7.36
	Par coupon - downturn recovery for ELD and average recovery for EL					
TTC	5.06	5.10	5.10	8.99	9.06	9.06
PIT	3.97	4.00	4.00	7.62	7.68	7.68
	Zero coupon - downturn recovery for ELD and average recovery for EL					
TTC	4.95	4.95	4.95	8.76	8.76	8.76
PIT	3.88	3.88	3.88	7.43	7.43	7.43
Panel C: 6% risk free rates						
	Par coupon - downturn recovery for both EL and ELD					
TTC	5.14	5.21	5.21	9.11	9.22	9.22
PIT	4.03	4.09	4.09	7.73	7.83	7.83
	Zero coupon - downturn recovery for both EL and ELD					
TTC	4.91	4.91	4.91	8.67	8.67	8.67
PIT	3.85	3.85	3.85	7.36	7.36	7.36
	Par coupon - downturn recovery for ELD and average recovery for EL					
TTC	5.18	5.25	5.25	9.21	9.32	9.33
PIT	4.06	4.11	4.12	7.80	7.90	7.91
	Zero coupon - downturn recovery for ELD and average recovery for EL					
TTC	4.95	4.95	4.95	8.76	8.76	8.76
PIT	3.88	3.88	3.88	7.43	7.43	7.43